

US chemist Wallace Carothers led the research team that invented neoprene, the world's first synthetic rubber, and nylon, the first fully synthetic fiber. His work altered the course of organic chemistry and ushered in the modern age of cheap plastics and other synthetic materials.

Born in Iowa, Wallace Hume Carothers was a chemistry instructor at Harvard University when, in 1928, the DuPont chemical company recruited him to investigate the chemical modification of natural materials to produce artificial ones. He studied polymers—substances found in all living organisms whose structure and chemical properties were not fully known. By 1929, Carothers believed that polymers were very large molecules made when smaller molecules join to form a long, repeating chain.

His idea was validated in the laboratory in 1930, when a test tube was found to contain a polymer similar to isoprene, found in natural rubber. The team had made a synthetic rubber, later dubbed neoprene. DuPont then tasked Carothers with finding a synthetic replacement for silk. In 1934, after experimenting with hundreds of chemical compositions, he found a formula for a super-strong, elastic polymer that could be drawn out into thin threads and woven into fine fabric: DuPont patented it as “nylon.”

“The **best organic chemist** in the **country.**”

Roger Adams

MILESTONES

JOINS DUPONT

Appointed director of research in organic chemistry in 1928, with brief to research polymers.

PUBLISHES RESEARCH

Writes articles on polymer theories in 1929 and 1931, establishing foundation of modern polymer science.

DISCOVERS NEOPRENE

Accidentally discovers artificial rubber with his team in 1930, using the polymerization process.

MAKES MIRACLE FIBER

Creates first human-made fiber, nylon, in 1934. Its first commercial use is in toothbrush bristles.



In May 1940, nylon stockings first went on sale in the US, and 800,000 pairs were bought. Demand continued unabated, and the name “nylons” became another word for stockings.

